SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission

media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A
Industry Problem Solving Task

Code of Conduct:

I D.Uday Kiran(192425161) certify that this submission is my original workand that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D.Uday Kiran

Industry Problem Solving Task

Network Topology Design Scenarios

1. Small Startup Office (20 Employees) — Star Topology

Problem Analysis

A 20-person startup requires a network that is simple, reliable, scalable, and easy to manage. Since each employee may use a computer or laptop, the network should also support printers, VoIP phones, wireless devices, and future expansion. Star topology is suitable because every device is connected to a central switch.

Solution Design

A 24-port managed switch is placed at the center of the network. Twenty employee computers, a printer, VoIP phones, and a wireless access point are connected to the switch. The switch is connected to a router or firewall for Internet access, DHCP, NAT, and security.

Technical Implementation

The network uses the following IP addressing scheme:

Network: 192.168.10.0/24
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Device	IP Address
Router	192.168.10.1
Switch	192.168.10.2
Access Point	192.168.10.3
PC1	192.168.10.10
PC2	192.168.10.11
PC3	192.168.10.12
PC4	192.168.10.13
PC5	192.168.10.14
PC6–PC20	192.168.10.15–29
Printer	192.168.10.30

Use of Tools & Technologies

- Cisco Packet Tracer
- Managed Layer 2/Layer 3 switch
- Router
- Firewall
- Wireless Access Point
- Cat6 UTP cable
- RJ-45 connectors
- IPv4 addressing
- DHCP
- NAT

Problem-Solving Approach

The network requirement is first identified based on the number of employees. A central switch is selected to provide connectivity to all devices. IP addresses are assigned systematically, and the router provides communication with external networks. The network is then tested using connectivity commands.

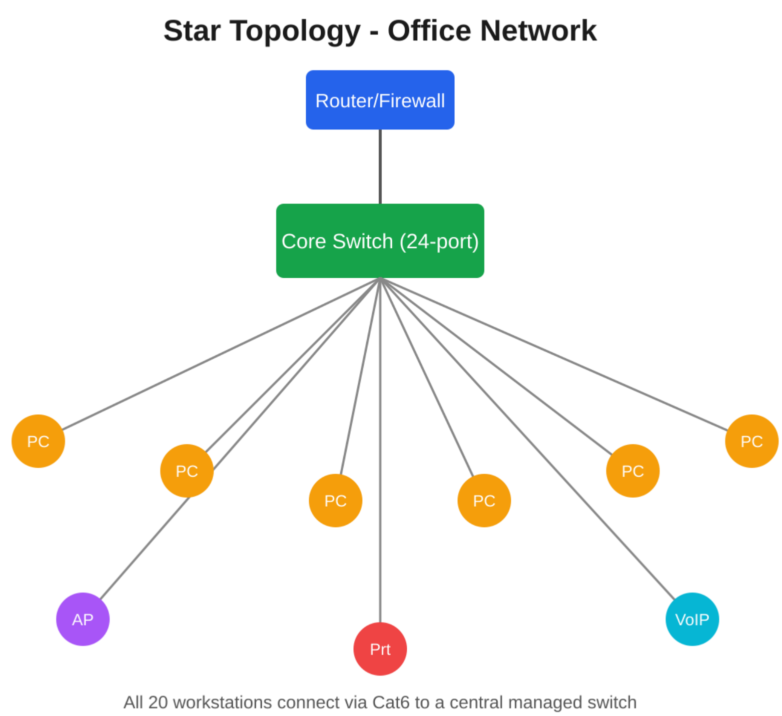

Figure 1: Star topology — all endpoints radiate from a central managed switch

2. Bank Requiring Near-Zero Downtime — (Partial) Mesh Topology

Problem Analysis

A bank requires very high network availability because network failure can affect financial transactions, ATMs, online banking, servers, and internal banking operations. Therefore, the network must provide redundant communication paths.

Solution Design

A partial mesh topology is recommended for the core network. Two enterprise routers, two core switches, and redundant firewalls are interconnected to provide alternative communication paths. Distribution and access switches are connected to multiple core switches.

Fiber-optic cables are used for connections between critical network devices because they provide high bandwidth, low latency, and immunity to electromagnetic interference.

Technical Implementation

The core network can use:

Network: 10.0.0.0/24
Subnet Mask: 255.255.255.0

Device	**IP Address**
Router 1	10.0.0.1
Router 2	10.0.0.2
Core Switch 1	10.0.0.10
Core Switch 2	10.0.0.11
Server 1	10.0.0.20
Server 2	10.0.0.21

OSPF can be configured for dynamic routing.

```
enable

configure terminal

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

exit

end

write memory
```

Use of Tools & Technologies

- Cisco Packet Tracer
- Enterprise routers
- Layer 3 switches
- Firewalls
- Fiber-optic cables
- OSPF
- BGP for external connectivity
- Redundant links
- UPS
- RAID storage

Problem-Solving Approach

The primary requirement is near-zero downtime. Redundant routers, switches, firewalls, and communication links are therefore selected. Dynamic routing protocols provide alternate paths when a link fails.

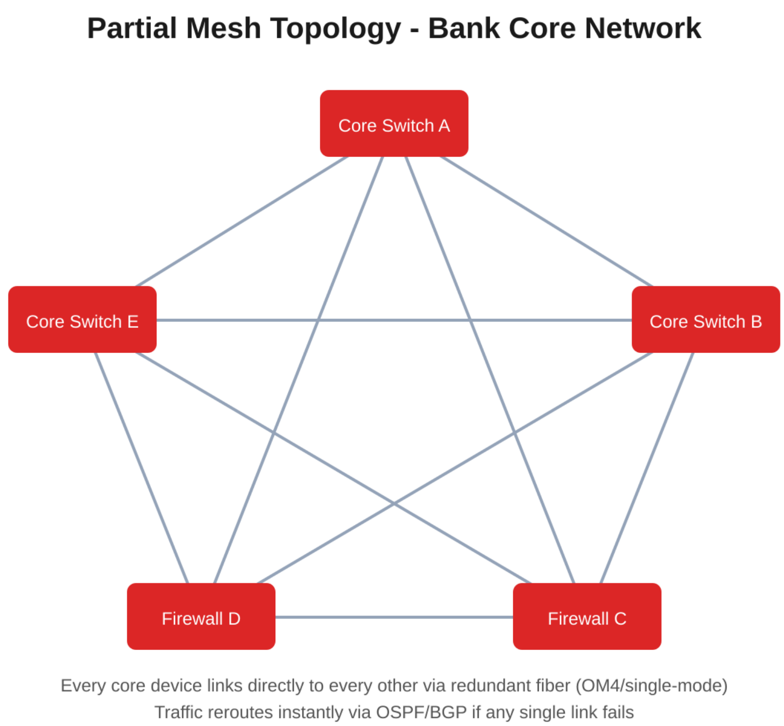

Figure 2: Partial mesh — every core device links directly to every other core device

3. Small Classroom Lab on a Tight Budget — Bus Topology

Problem Analysis

A small classroom laboratory with limited financial resources requires a low-cost network. A traditional bus topology uses a single backbone cable shared by all computers, reducing the amount of networking equipment required.

Solution Design

The traditional bus network uses a common coaxial backbone cable. Computers are connected to the backbone using T-connectors and BNC connectors. A 50-ohm terminator is placed at both ends of the backbone to prevent signal reflection.

Technical Implementation

The classroom network can use:

Network: 192.168.20.0/24
Subnet Mask: 255.255.255.0

Device	IP Address
PC1	192.168.20.10
PC2	192.168.20.11
PC3	192.168.20.12
PC4	192.168.20.13
PC5	192.168.20.14

Traditional Ethernet bus networks use RG-58 coaxial cable, BNC connectors, T-connectors, and 50-ohm terminators.

Use of Tools & Technologies

- Coaxial cable
- RG-58 cable
- BNC connectors
- T-connectors
- 50-ohm terminators
- Network interface cards
- Cisco Packet Tracer for conceptual representation

Problem-Solving Approach

The main requirement is low cost. A shared backbone is selected to reduce networking hardware. IP addresses are assigned to the computers and connectivity is tested.

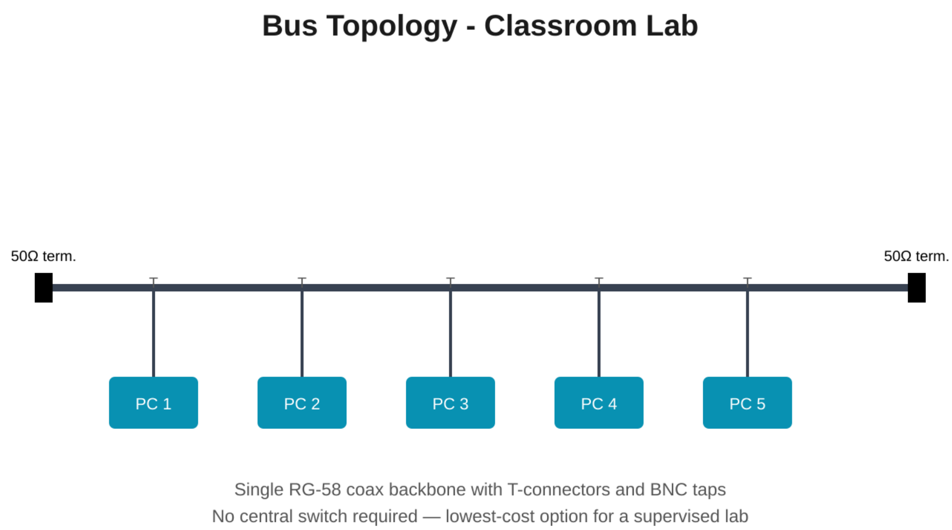

Figure 3: Bus topology — a single shared backbone terminated at both ends

4. Large University Campus — Hybrid Topology

Problem Analysis

A large university campus contains multiple buildings such as lecture halls, laboratories, libraries, hostels, and administrative offices. Thousands of users require reliable, scalable, and high-speed network connectivity.

A single topology cannot efficiently satisfy all campus requirements. Therefore, a hybrid topology is appropriate.

Solution Design

The campus backbone uses a fiber-based ring or partial mesh for redundancy. Each building uses a star topology, where a distribution switch connects to access switches. End devices such as computers, printers, IP phones, and wireless access points are connected to access switches.

Technical Implementation

The campus can use the network:

Network: 10.10.0.0/16

Building	Network
Computing	10.10.10.0/24
Electronics	10.10.20.0/24
Mechanical	10.10.30.0/24

Building	Network
Library	10.10.40.0/24
Administration	10.10.50.0/24

Example core router configuration:

```
enable
configure terminal
hostname CAMPUS-CORE

interface gigabitEthernet 0/1
ip address 10.10.0.1 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/2
ip address 10.10.0.5 255.255.255.252
no shutdown
exit

router ospf 1
network 10.10.0.0 0.0.255.255 area 0
exit

end
write memory
```

Use of Tools & Technologies

- Cisco Packet Tracer
- Core Layer 3 switches
- Distribution switches
- Access switches
- Routers
- Fiber-optic cables
- Cat6 UTP cables
- Wireless access points
- PoE
- OSPF

Problem-Solving Approach

The campus is divided into core, distribution, and access layers. Fiber is selected for the high-speed backbone, while Cat6 is used for connections within buildings. Redundant backbone links provide reliability.

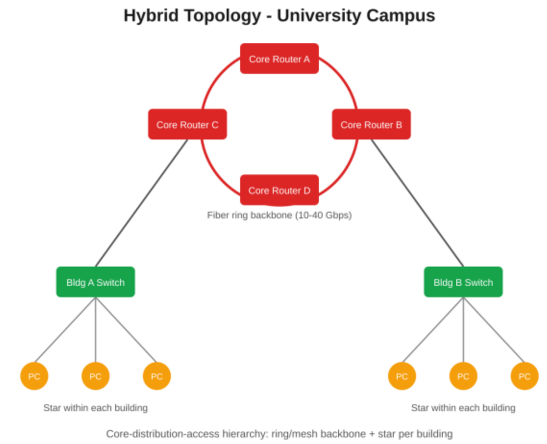

Figure 4: Hybrid topology — ring/mesh backbone with a star inside each building

5. Multi-Department Company — Hierarchical VLAN Design

Problem Analysis

A company with HR, Finance, Sales, and IT departments requires secure and organized communication. If all departments use the same network, broadcast traffic increases and sensitive information may be exposed.

VLANs provide logical separation between departments.

Solution Design

Each department is assigned a separate VLAN. Layer 2 access switches connect employee devices, while a Layer 3 switch performs inter-VLAN routing. A router or firewall provides Internet connectivity and security.

Technical Implementation

Department	VLAN	Network
HR	10	192.168.10.0/24
Finance	20	192.168.20.0/24
Sales	30	192.168.30.0/24
IT	40	192.168.40.0/24

VLAN configuration:

```
enable
configure terminal

vlan 10
name HR
```

```
exit

vlan 20
name FINANCE
exit

vlan 30
name SALES
exit

vlan 40
name IT
exit
```

HR access ports:

```
interface range fastEthernet 0/1-5
switchport mode access
switchport access vlan 10
exit
```

Finance access ports:

```
interface range fastEthernet 0/6-10
switchport mode access
switchport access vlan 20
exit
```

Sales access ports:

```
interface range fastEthernet 0/11-15
switchport mode access
switchport access vlan 30
exit
```

IT access ports:

```
interface range fastEthernet 0/16-20
switchport mode access
switchport access vlan 40
exit
```

Trunk configuration:

```
interface gigabitEthernet 0/1
switchport mode trunk
exit
```

Inter-VLAN routing:

```
enable
configure terminal
ip routing
```

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit

end
write memory
```

Use of Tools & Technologies

- Cisco Packet Tracer
- Layer 2 switches
- Layer 3 switch
- Router
- Firewall
- VLAN
- Inter-VLAN routing
- Cat6 UTP
- IPv4 addressing
- Access Control Lists

Problem-Solving Approach

The company is divided into logical departments. Each department receives a separate VLAN and IP subnet. A Layer 3 switch provides controlled communication between VLANs, while security policies can be implemented using ACLs.

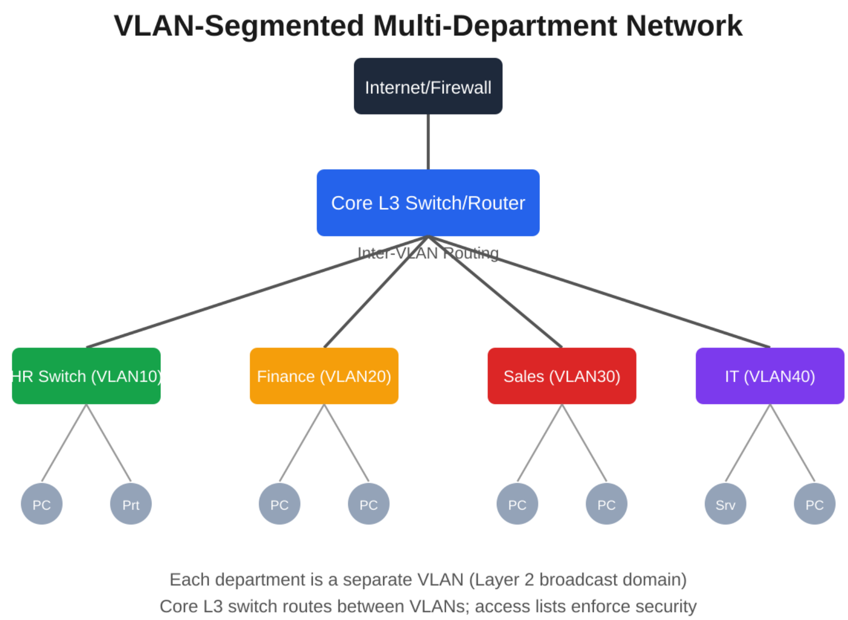

Figure 5: VLAN-segmented network — one VLAN per department, routed centrally